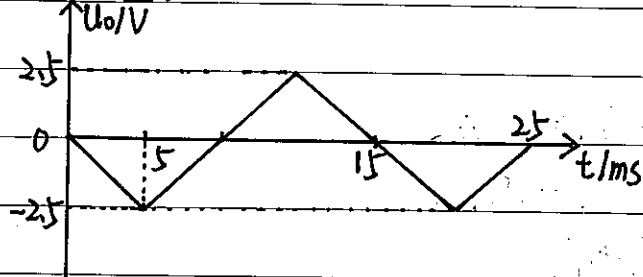


$$\therefore U_0 = 10(U_{I1} + U_{I2} + U_{I3})$$

$$6.11 \quad U_0 = -\frac{1}{RC} \int u_i dt = -100 \int u_i dt$$

$\therefore$ 如图



6.13 (a) 由“虚短”和“虚断”

$$\frac{U_I - 0}{R_1} = \frac{0 - (U_C + U_0)}{R_2} = \dot{U}_C = \dot{U}_R$$

$$\begin{aligned} U_0 &= -U_C - U_{0I} = -\frac{1}{C} \int \dot{U}_C dt - U_I \\ &= -\frac{1}{RC} \int U_I dt - U_I = -U_I - 1000 \int U_I dt \end{aligned}$$

(b) 由“虚短”和“虚断”

$$\begin{cases} U_{C1} = U_{I1} - U_- = U_{I1} = \frac{1}{C_1} \int \dot{U}_{C1} dt \\ \dot{U}_{C1} = \dot{U}_R = \dot{U}_{C2} = \frac{0 - (U_{C2} + U_0)}{R} \end{cases}$$

$$\begin{aligned} \therefore U_0 &= -R\dot{U}_{C1} - U_{C2} = -R\dot{U}_{C1} - \frac{1}{C_2} \int \dot{U}_{C2} dt \\ &= -R\dot{U}_{C1} - \frac{1}{C_2} \int \dot{U}_{C1} dt \\ &= -RC_1 \frac{dU_{I1}}{dt} - \frac{C_1}{RC_2} U_{I1} \\ &= -10^{-3} \frac{dU_I}{dt} - 2U_I \end{aligned}$$

(c). 由“虚短”和“虚短断”

$$\begin{cases} \frac{U_- - U_+}{R} = \dot{U}_{C1} = C \frac{dU_{C1}}{dt} = C \frac{d(U_0 - U_+)}{dt} \\ \frac{U_I - U_+}{R} = \dot{U}_{C2} = C \frac{dU_{C2}}{dt} = C \frac{dU_+}{dt} \end{cases}$$